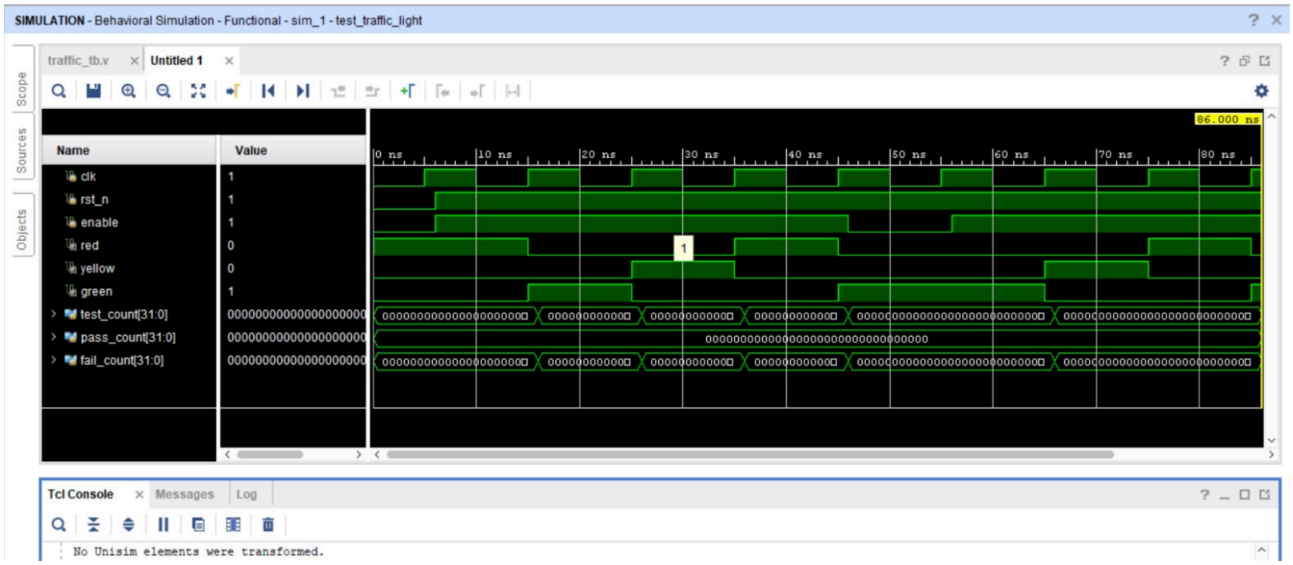
```
``timescale 1ns/1ps

module traffic_light(

input clk,
input rst_n,
input enable,

output reg red,
output reg yellow,
output reg green

);

parameter RED    = 2'b00;
parameter GREEN  = 2'b01;
parameter YELLOW = 2'b10;

reg [1:0] state;

//
// State register with synchronous enable
//
always @(posedge clk or negedge rst_n)
begin
   if(!rst_n)
      state <= RED;
   else if(enable)
   begin
      case(state)
         RED:    state <= GREEN;
         GREEN:  state <= YELLOW;
         YELLOW: state <= RED;
         default:state <= RED;
      endcase
   end
end

//
// Output logic (Moore FSM)
//
always @(state)
begin
   case(state)

      RED:
      begin
         red    = 1;
         yellow = 0;
         green  = 0;
      end

      GREEN:
      begin
         red    = 0;
         yellow = 0;
         green  = 1;
      end

      YELLOW:
```

```
        begin
            red    = 0;
            yellow = 1;
            green  = 0;
        end

        default:
        begin
            red    = 1;
            yellow = 0;
            green  = 0;
        end

    endcase
end

endmodule

`timescale 1ns/1ps

module test_traffic_light;

    reg clk, rst_n, enable;
    wire red, yellow, green;

    traffic_light dut (
        .clk(clk),
        .rst_n(rst_n),
        .enable(enable),
        .red(red),
        .yellow(yellow),
        .green(green)
    );

    // CLOCK
    initial clk = 0;
    always #5 clk = ~clk;

    integer pass = 0;
    integer fail = 0;

    // TASK TO CHECK OUTPUT
    task check;
        input er, ey, eg;
        begin
            @(posedge clk); #1;
            if (red==er && yellow==ey && green==eg) begin
                $display("PASS -> R=%b Y=%b G=%b", red, yellow, green);
                pass++;
            end else begin
                $display("FAIL -> R=%b Y=%b G=%b", red, yellow, green);
                fail++;
            end
        end
    endtask

    initial begin
        $dumpfile("traffic_light.vcd");
        $dumpvars(0,test_traffic_light);
```

```
        $display("==== TRAFFIC LIGHT TEST START ====");

        // INITIAL VALUES
        enable = 0;
        rst_n  = 0;

        // APPLY RESET
        repeat(2) @(posedge clk);
        rst_n = 1;

        // AFTER RESET → MUST BE RED
        check(1,0,0);

        // ENABLE STATE TRANSITIONS
        enable = 1;

        check(0,0,1); // GREEN
        check(0,1,0); // YELLOW
        check(1,0,0); // RED

        // DISABLE ENABLE → STATE MUST HOLD
        enable = 0;
        @(posedge clk);
        check(1,0,0);

        $display("================================");
        $display("PASS = %0d, FAIL = %0d", pass, fail);

        if (fail == 0)
            $display("🎉 ALL TESTS PASSED");
        else
            $display("❌ SOME TESTS FAILED");

        #20 $finish;
    end

endmodule
```